

⇒ Rocket Traj. Sim. Exercise 2 Notes:

Modify exercise 1 to apply constant gravity & a thrust cutoff when propellant is expended

> Constant acceleration due to gravity given by:

$$a_g = \frac{GM_{\oplus}}{R_{\oplus}^2} \approx 9.81 = g_0$$

where

$$G = 6.6743 \times 10^{-11} \text{ [Nm}^2\text{/kg}^2\text{]}$$

$$M_{\oplus} = 5.9722 \times 10^{24} \text{ [kg]}$$

$$R_{\oplus} = 6.371 \times 10^6 \text{ [m]}$$

> Thrust cutoff occurs when propellant expended (i.e. "burnout"):

$$m_{\text{prop}} - \dot{m}t = 0$$

where

m_{prop} = propellant mass [kg]

$\dot{m}$ = propellant mass flow rate [kg/s]

t = time after rocket engine ignition [s]

Solve for time;

$$\dot{m}t = m_{\text{prop}}$$

$$t = \frac{m_{\text{prop}}}{\dot{m}}$$

is time at which burnout occurs